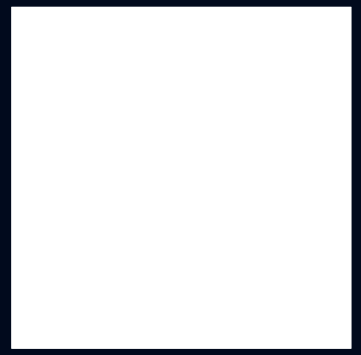


Ambient Systems

Climate Management for Cities of Tomorrow

Team

Ivan Damnjanovic
VP Product Development

TBC
VP Sales & Marketing

TBC
VP Engineering

WIP – start with indoor farms?

Modern Buildings
Face Many Challenges

1. Climate Change
2. Urban Pollution
3. Indoor Contaminants
4. Rising Energy Costs
5. Increasing Technological Complexity



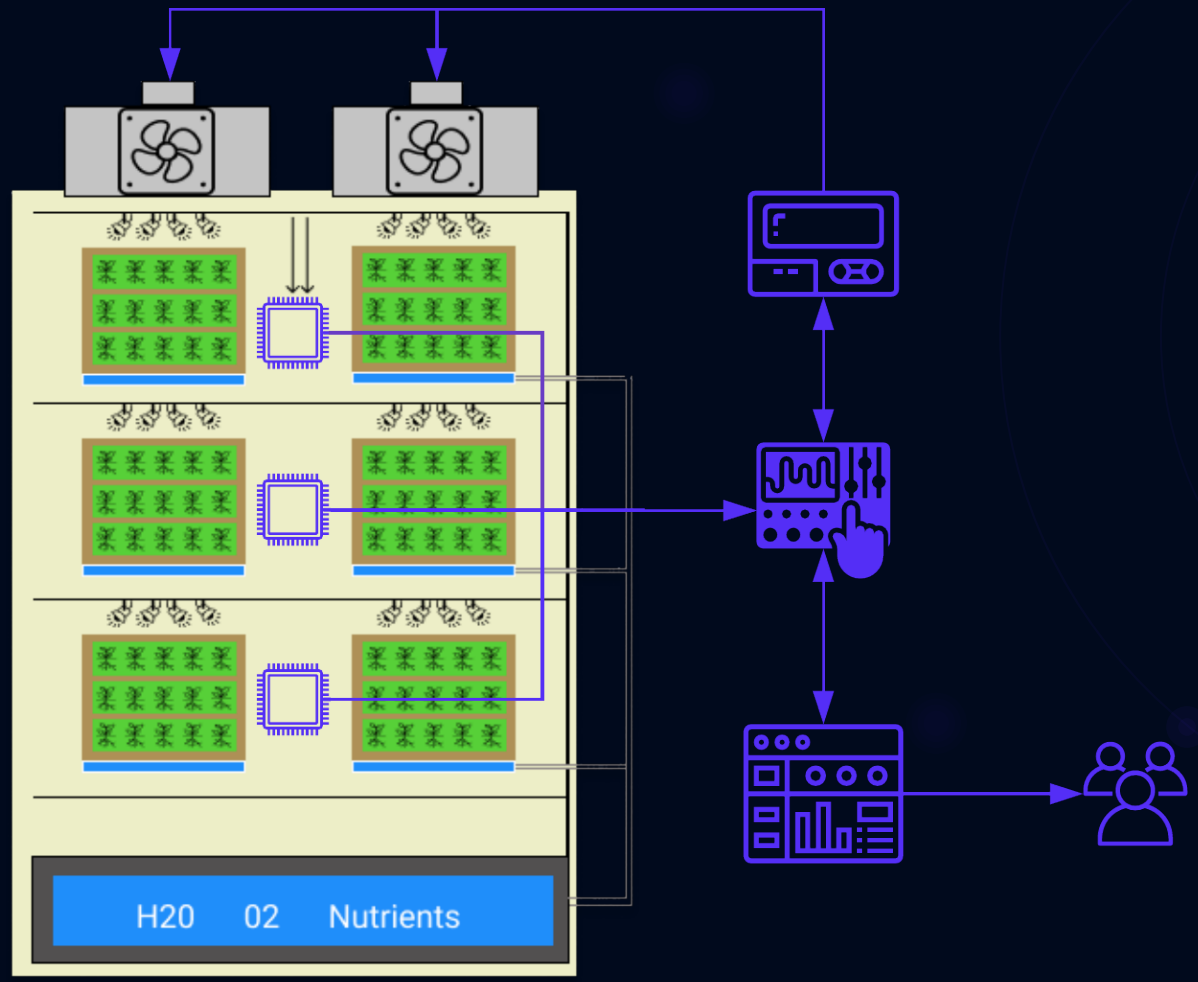
Indoor Farm Opportunity

1. Vertical Farming is a rapidly growing industry with 23.62% CAGR*.

2. Asia-Pacific is the global leader in vertical farming with market value of $355M*.

3. Clear need from several farm operators in Singapore.

*Source: Statista



Solution

4-Part Tech Stack

Combine Ambient Sensors, [redacted], and Controller with the exiting equipment to provide an additional layer of intelligence for building optimization.

Connect

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Monitor

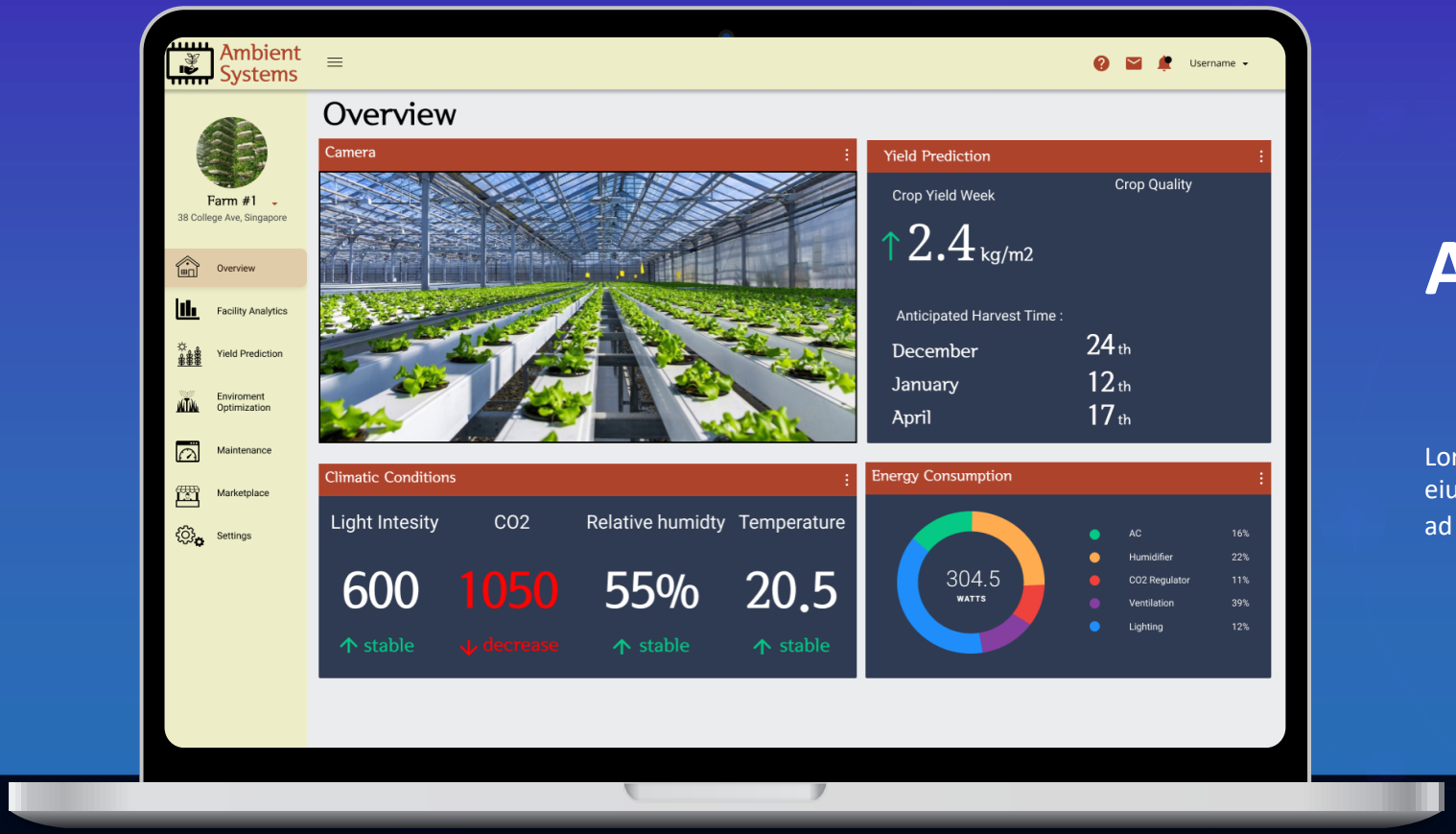
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Ambient Platform

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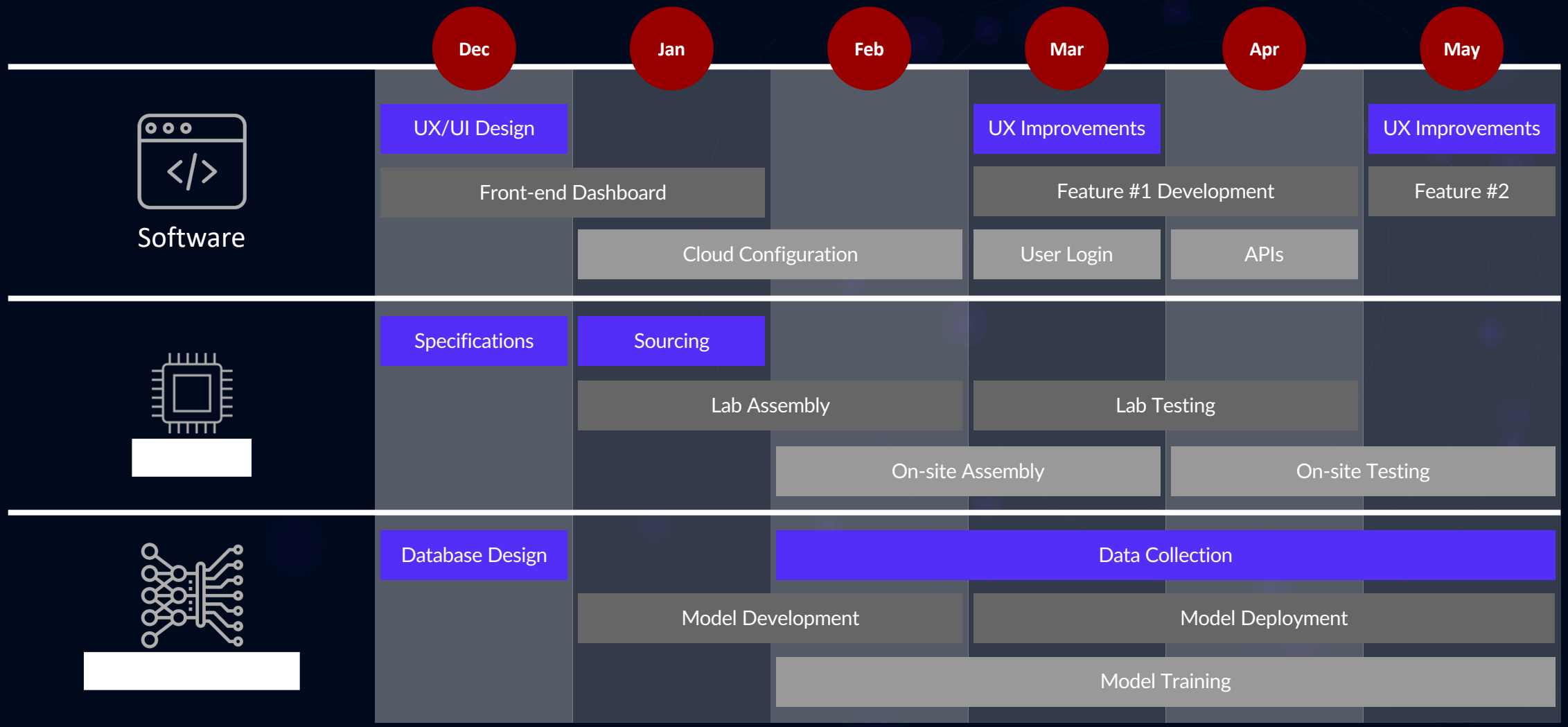
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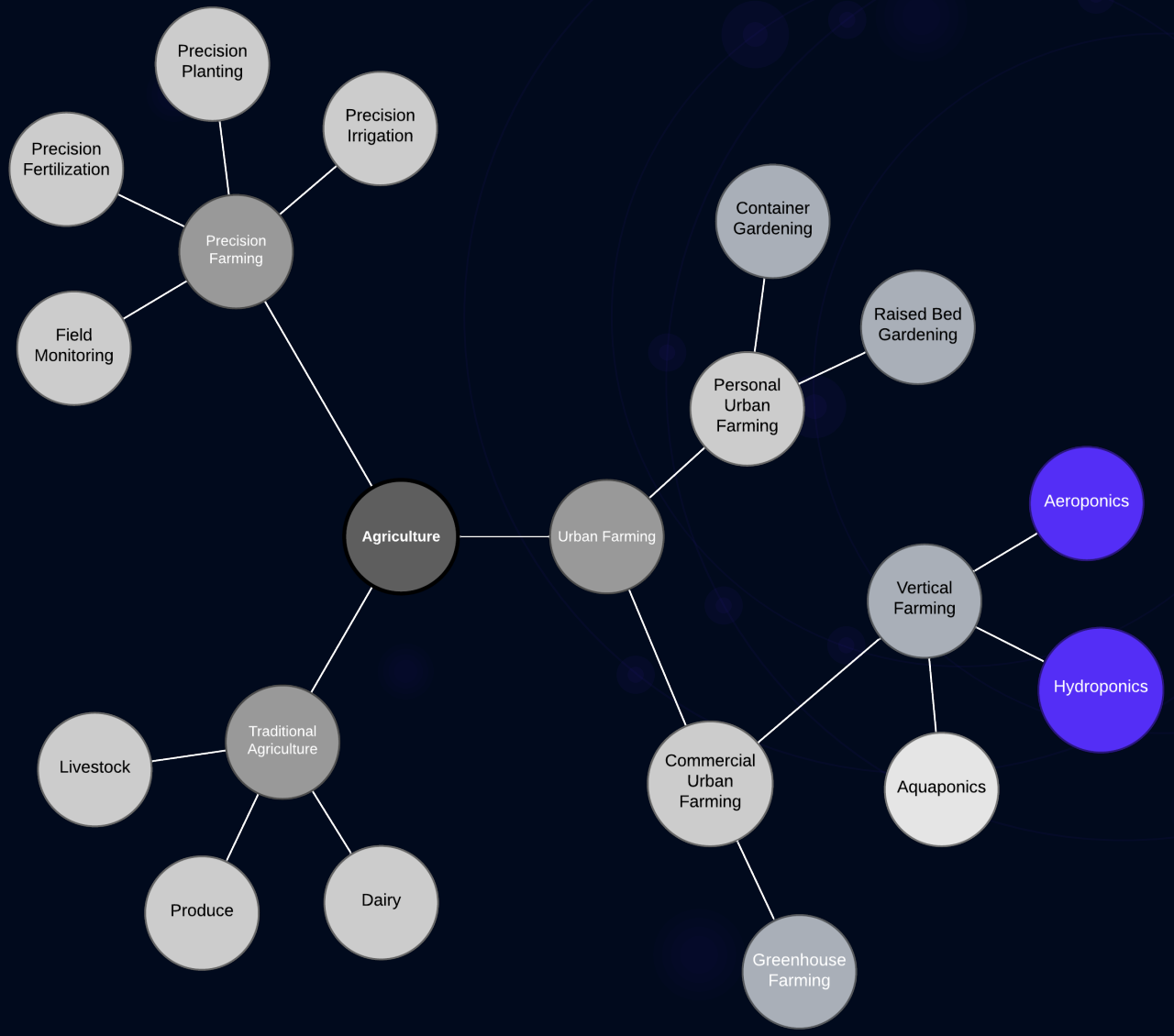
PRODUCT ROADMAP

WIP – move to milestones?



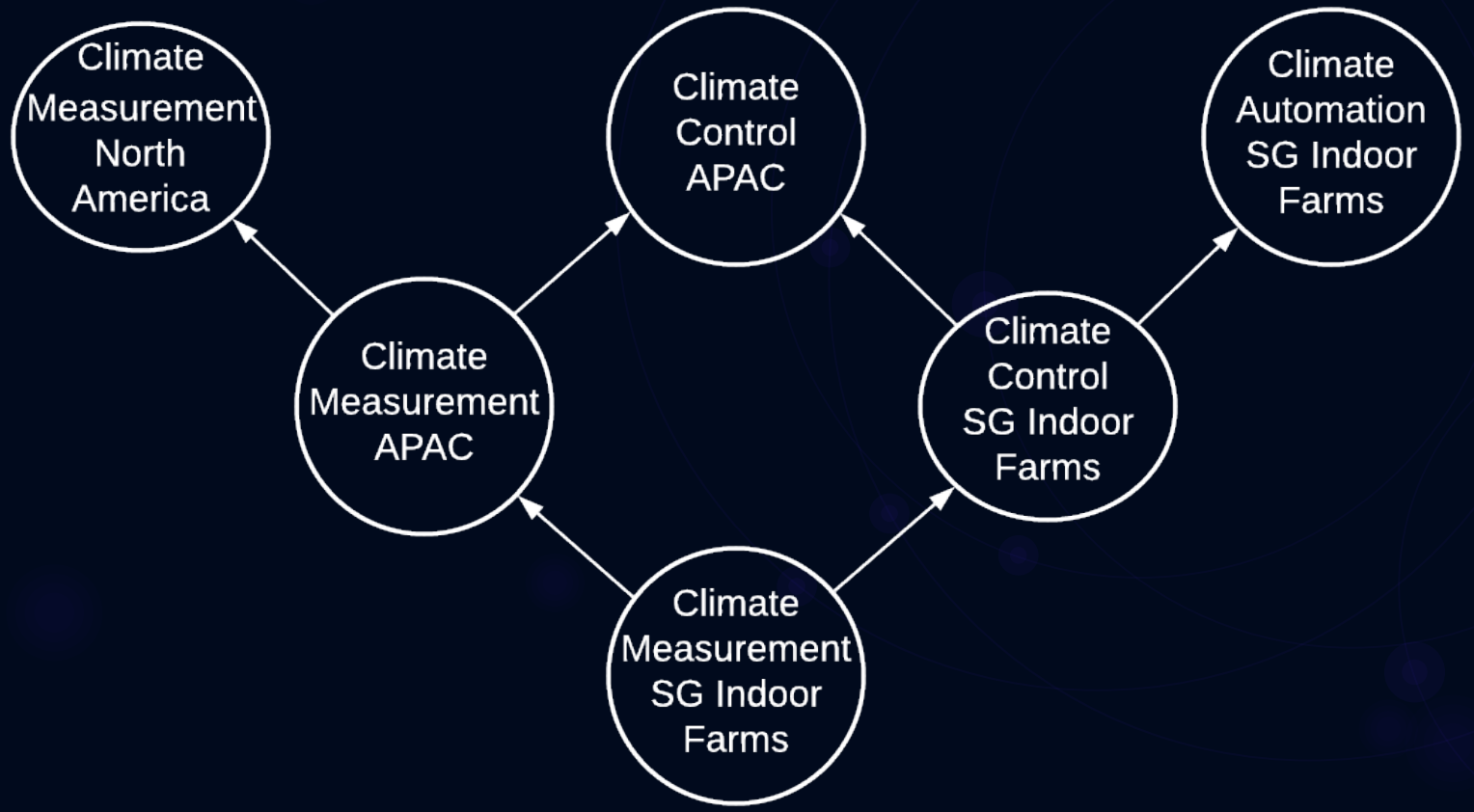
Market: Segmentation

WIP – Simplify and increase size



Growth: Beachhead market

WIP – Add market size for each segment



Adjacent Sectors

Each building has unique operational needs, yet property owners and facility managers are short on in-house skilled talent that can manage and optimize their existing building systems in real-time.



Indoor Farms

Increase crop yield and streamline operations

Offices

Increase worker comfort and productivity

Historical Buildings & Museums

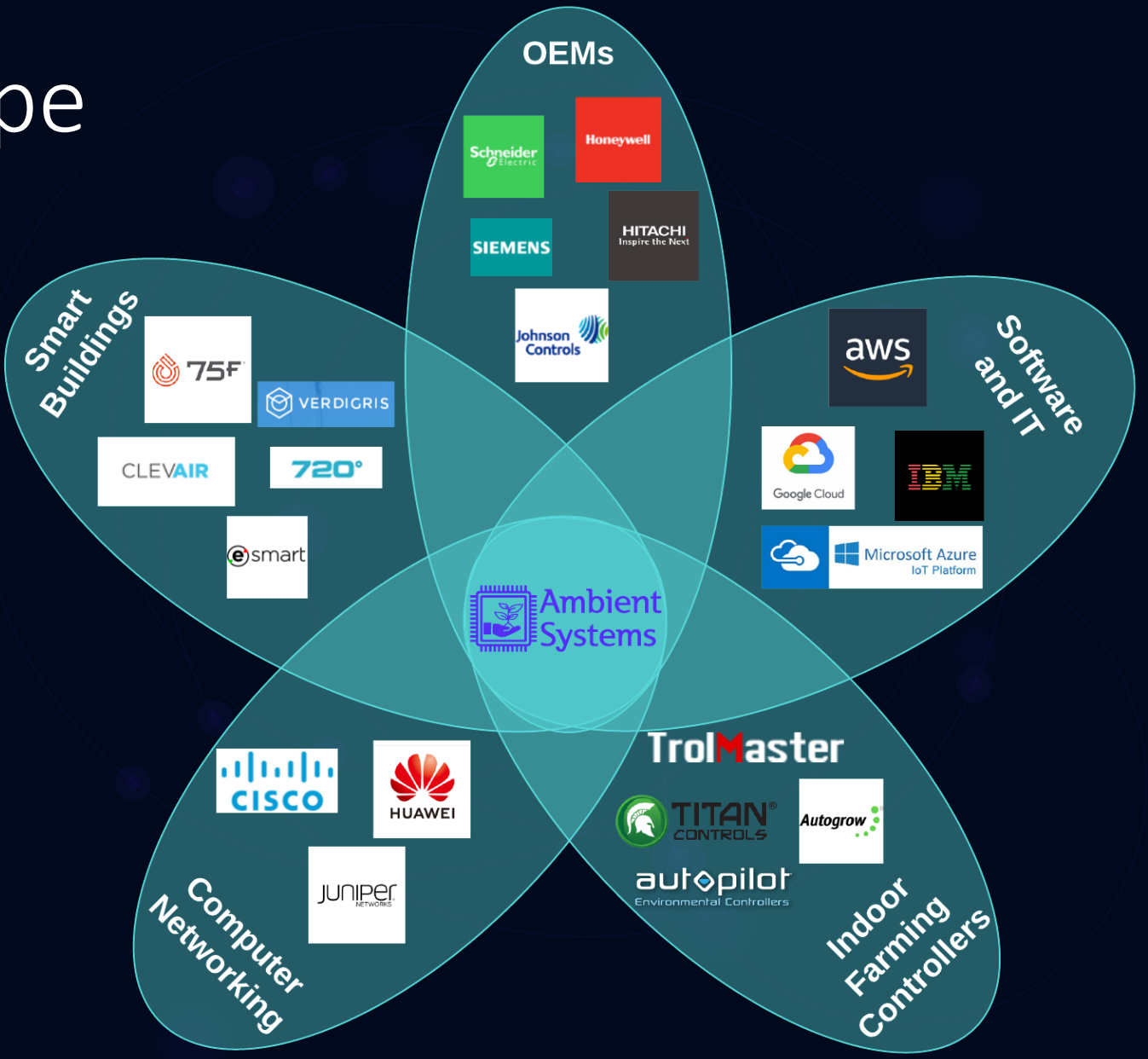
Preserve interiors of historical buildings and protect invaluable artifacts

Healthcare

Improve patient outcomes and minimize airborne infections

Competitive Landscape

Ambient is positioned at the intersection of multiple segments in order to provide a tailored solution for the beachhead market: indoor farms.



Competitive Benchmark

	Ambient	Autogrow	75F	Johnson Controls
Segment	Multiple	Indoor Farming Controller	Smart Buildings	OEM
Proximity to Customer (Low Logistics Cost)	✓ Yes	✗ No	✗ No	✓ Yes
Specific vs. Generic	✓ Indoor Farms	✓ Indoor Farms	✗ No	✗ No
Modern Web-based UI	✓ Yes	✗ No	✓ Yes	✗ Some
Wireless IoT Network	✓ Yes	✗ No	✓ Yes	✓ Yes
Advanced Optimization Methods	✓ ML models	✗ No	✗ Some	✗ Some
Pricing	Quotation (based on area)	$3000 for one unit + Installation Fees	Quotation	Quotation

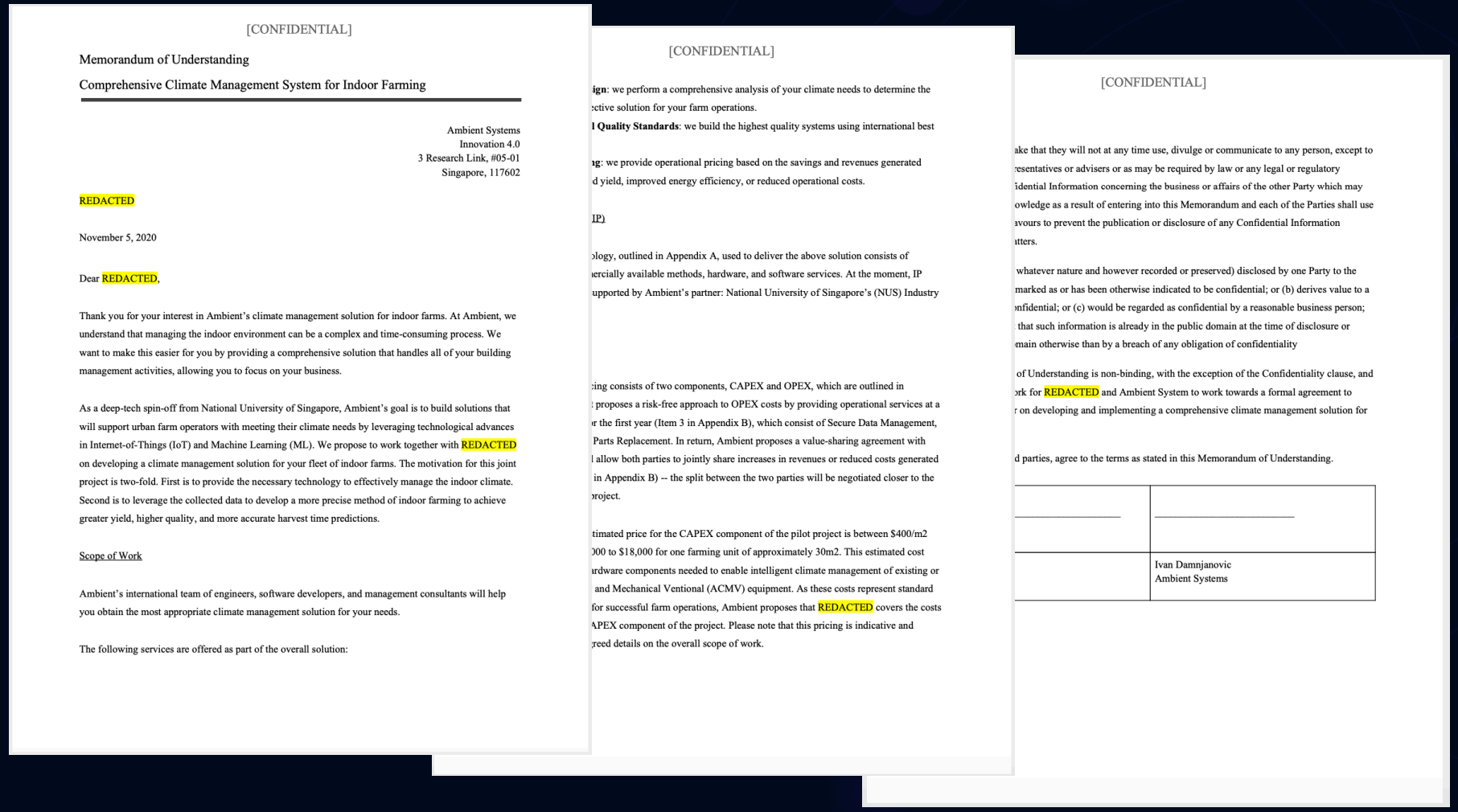
Innovative Revenue Model

Section	Component	Cost
CAPEX		
Services	Audit and Design	Complimentary
	Installation	$4,000 - $5,000
Hardware	Ambient Smart Sensors	$1,500 - $2,500
	Ambient Edge Node	$2,500 - $4,500
	Ambient Control Unit	$4,000 - $6,000
	CAPEX SubTotal	**$12,000 - $18,000**
OPEX		
Operations	Secure Data Management	Complimentary
	System Maintenance	Complimentary
	Parts Replacement	Complimentary
Ambient Platform	Facility Analytics	Value-sharing
	ACMV Optimization	Value-sharing
	Predictive Maintenance	Value-sharing
	Yield Prediction	Value-sharing
	Marketplace	Value-sharing
	OPEX SubTotal	**$0**
	Estimated Total Cost	**$12,000 - $18,000 + OPEX value-sharing**

WIP – , Funding Gap, Valuation

Financials

Traction: LOI/MOU



[CONFIDENTIAL]

Memorandum of Understanding

Comprehensive Climate Management System for Indoor Farming

Ambient Systems
Innovation 4.0
3 Research Link, #05-01
Singapore, 117602

REDACTED

November 5, 2020

Dear REDACTED,

Thank you for your interest in Ambient's climate management solution for indoor farms. At Ambient, we understand that managing the indoor environment can be a complex and time-consuming process. We want to make this easier for you by providing a comprehensive solution that handles all of your building management activities, allowing you to focus on your business.

As a deep-tech spin-off from National University of Singapore, Ambient's goal is to build solutions that will support urban farm operators with meeting their climate needs by leveraging technological advances in Internet-of-Things (IoT) and Machine Learning (ML). We propose to work together with REDACTED on developing a climate management solution for your fleet of indoor farms. The motivation for this joint project is two-fold. First is to provide the necessary technology to effectively manage the indoor climate. Second is to leverage the collected data to develop a more precise method of indoor farming to achieve greater yield, higher quality, and more accurate harvest time predictions.

Scope of Work

Ambient's international team of engineers, software developers, and management consultants will help you obtain the most appropriate climate management solution for your needs.

The following services are offered as part of the overall solution:

[CONFIDENTIAL]

ign: we perform a comprehensive analysis of your climate needs to determine the ective solution for your farm operations.
l Quality Standards: we build the highest quality systems using international best

ng: we provide operational pricing based on the savings and revenues generated d yield, improved energy efficiency, or reduced operational costs.

IP)

ology, outlined in Appendix A, used to deliver the above solution consists of ercially available methods, hardware, and software services. At the moment, IP upported by Ambient's partner: National University of Singapore's (NUS) Industry

cing consists of two components, CAPEX and OPEX, which are outlined in proposes a risk-free approach to OPEX costs by providing operational services at a r the first year (Item 3 in Appendix B), which consist of Secure Data Management, Parts Replacement. In return, Ambient proposes a value-sharing agreement with l allow both parties to jointly share increases in revenues or reduced costs generated in Appendix B) -- the split between the two parties will be negotiated closer to the project.

timated price for the CAPEX component of the pilot project is between $400/m2 000 to $18,000 for one farming unit of approximately 30m2. This estimated cost ardware components needed to enable intelligent climate management of existing or and Mechanical Ventional (ACMV) equipment. As these costs represent standard for successful farm operations, Ambient proposes that REDACTED covers the costs APEX component of the project. Please note that this pricing is indicative and reed details on the overall scope of work.

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ake that they will not at any time use, divulge or communicate to any person, except to resentatives or advisers or as may be required by law or any legal or regulatory idential Information concerning the business or affairs of the other Party which may owledge as a result of entering into this Memorandum and each of the Parties shall use avours to prevent the publication or disclosure of any Confidential Information tters.

whatever nature and however recorded or preserved) disclosed by one Party to the marked as or has been otherwise indicated to be confidential; or (b) derives value to a nfidential; or (c) would be regarded as confidential by a reasonable business person; that such information is already in the public domain at the time of disclosure or main otherwise than by a breach of any obligation of confidentiality

of Understanding is non-binding, with the exception of the Confidentiality clause, and rk for REDACTED and Ambient System to work towards a formal agreement to r on developing and implementing a comprehensive climate management solution for

d parties, agree to the terms as stated in this Memorandum of Understanding.

_____	_____
	Ivan Damnjanovic Ambient Systems

Summary

1. Vertical farming is a high-growth and high-value market.
2. Ambient provides a solution for climate management using IoT and AI.
3. Pilot site planned with a local indoor farm.